Article

Theory-Based Interpretability in Deep Knowledge Tracing via Grounded Transformers

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Abstract

Knowledge Tracing enables estimation of how students' knowledge evolves through interactions with educational content, forming a cornerstone of intelligent tutoring systems. While deep learning models achieve superior predictive performance, they lack interpretability which is problematic in educational contexts where transparency and pedagogical grounding are essential. We introduce gTransformer, a Transformer-based model bridging deep learning performance with intrinsic interpretability through Representational Grounding. The architecture enriches student interaction sequences with Bayesian Knowledge Tracing (BKT) parameters. Attention mechanisms integrate these embeddings into latent representations, projected into individualized learning parameters constrained to preserve theoretical semantics. These drive predictions through interpretable BKT logic, enabling theory-grounded explanations. Validation demonstrates: (1) structural encoding organizing around BKT constructs (probing selectivity $\Delta R^2 > 0.65$); (2) semantic alignment preserving pedagogical meaning ($\rho_T = 0.604$); (3) functional alignment with quantified confidence. Interpretable predictions surpass classical BKT (+9.0% AUC) with minimal cost versus black-box architectures. Critically, gTransformer enables context-aware personalization by differentiating students based on longitudinal learning trajectories rather than on current responses—capturing patterns Markovian models cannot represent. This offers a practical path toward AI-driven personalization that is both accurate and interpretable, supporting educators in tailoring adaptive instruction.

Keywords: deep knowledge tracing; theory-based interpretability; grounded transformers; bayesian knowledge tracing; educational data analysis; personalized learning

1. Introduction

Knowledge Tracing (KT) [1] is a foundational problem in Intelligent Tutoring Systems and Artificial Intelligence in Education. It is formally defined as the supervised sequence modeling task of estimating a student's evolving mastery based on their historical interactions. The input to the system consists of a temporal sequence $\mathcal{S} = \{(q_1, r_1), (q_2, r_2), \ldots, (q_{t-1}, r_{t-1})\}$, where each tuple represents a question (or skill) q_i and the student's binary response $r_i \in \{0, 1\}$. The task objective is to predict the output $\hat{y}_t = P(r_t = 1 | q_t, \mathcal{S})$, signifying the probability of a correct response to the current question q_t given the observed history. By monitoring these longitudinal estimations, we can track the dynamic evolution of student knowledge, providing a data-driven foundation for

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personalized instruction and targeted pedagogical interventions. In an era where educational environments are becoming increasingly digital and diverse, the ability to accurately track and interpret student mastery is not merely a technical optimization but a critical requirement for effective, scalable, and learner-centred education.

Historically, the development of KT has been characterized by a dichotomy between two distinct paradigms [2]. The first, which predominated in the field for years, is characterized by probabilistic models such as Bayesian Knowledge Tracing (BKT) [1] and Factor Analysis Models based on logistic regression, including Additive Factor Models [3], Performance Factor Analysis [4], and Knowledge Tracing Machines [5]. Factor Analysis Models are theoretically supported by the Item Response Theory (IRT) [6], which has played a large role in educational assessment and measurement. These traditional approaches are intrinsically interpretable: they model learning as a process governed by explicit parameters related to pedagogical, cognitive or psychometric principles. Because the model structure mirrors a theoretical understanding of human learning, educators can inspect these values to validate the model's reasoning and trust its outputs. However, this transparency comes at the cost of representational rigidity; the simplifying assumptions required by these models often limit their predictive power, causing them to struggle with the complex, non-linear, and long-range dependencies inherent in real-world student behavior [7].

The second paradigm emerged with the advent of Deep Knowledge Tracing (DKT) [8], which leverages deep learning techniques ranging from initial Recurrent Neural Networks to more recent Transformer architectures [9]. The latter, based on encoder-decoder components and attention mechanisms, are especially relevant as they provide the foundation for current Large Language Models. By treating student interaction histories as temporal sequences, Transformer models apply techniques similar to those that characterize LLMs. DKT models have achieved state-of-the-art predictive performance, significantly outperforming classical baselines due to a structural high capacity that allow them to extract latent features from vast amounts of data [2]. However, this substantial gain in predictive performance has come at the cost of model transparency, creating a significant interpretability gap. Deep learning models are notoriously opaque "black boxes", where the internal high-dimensional representations bear no direct correspondence to constructs that are interpretable in terms of human-understandable concepts or established domain knowledge. In a high-stakes domain such as education, where algorithmic decisions influence learning trajectories, grades, and opportunities—this lack of transparency poses critical challenges for practical adoption.

Extensive research has sought to bridge this gap, with Bai et al. [10] classifying modalities into *post-hoc* and *ante-hoc* categories. While *post-hoc* methods aim to extract explanations from black-box models after training, *ante-hoc* approaches seek to achieve intrinsic interpretability by design, ensuring the model's internal logic is transparent from the outset. Within this latter category, recent models have attempted to achieve intrinsic transparency by integrating attention mechanisms [11,12] or by incorporating side information such as Item Response Theory (IRT) parameters [13–15]. These efforts aim to make the decision-making process visible, moving away from pure black-box architectures toward systems whose internal states can be probed or visualized.

However, from the perspective of domain experts such as teachers and pedagogical designers, these solutions frequently fall short. Post-hoc explanations often rely on technical relevance scores or local approximations that remain disconnected from established pedagogical principles [16] while most current ante-hoc methods do not produce outputs that map directly to human-understandable educational constructs. As noted in various surveys [10,17], when deep learning models incorporate domain knowledge, the primary objective is typically the optimization of predictive performance, while interpretability is often overlooked or left as an unaddressed concern.

Consequently, there is a critical need for new approaches that move beyond technical transparency toward true pedagogical interpretability. For a KT system to be confidently adopted in high-stakes educational environments, its estimations must be grounded in trusted and empirically validated principles derived from relevant fields such as pedagogy, cognitive science, or psychometrics. Educators require models that speak their language, tracking mastery not through opaque latent features, but through concepts explicitly defined by established frameworks, including the skills, prior knowledge, or learning rates postulated by traditional approaches such as BKT and Factor Analysis Models.

To address this challenge, we introduce the *grounded Transformer (gTransformer)*, a variant of attention-based Transformers that implements a novel *Latent Grounding* methodology to bridge the gap between deep learning expressiveness and theoretical alignment. This approach explicitly anchors the Transformer's latent representations to the semantic constructs of an intrinsically interpretable reference model. The gTransformer architecture utilizes an encoder-decoder design with attention mechanisms to create context vectors that encode complex student interaction patterns. These vectors are projected into enriched parameters, which then drive predictions using the reference model's interpretable logic. By doing so, the model's estimations can be tracked down to domain knowledge and theoretical principles. This architecture satisfies the requirements for intrinsic interpretability by providing "grounded interpretability" by design, as the estimations are expressed through the same well-validated concepts used in established interpretable systems.

Our approach contributes to bridging the dichotomy between high-capacity deep learning and domain-specific theoretical transparency, providing two practical advantages. First, compared to non-interpretable DKT models, gTransformer offers intrinsic pedagogical transparency by generating actionable parameters, such as prior knowledge and learning rates, that are directly meaningful to educators. Second, compared to traditional approaches, gTransformer is context-aware, i.e it is able to encode individual patterns taking into account the history of student interactions, overcoming the limitations of rigid population-level parameters and Markovian assumptions characteristic of BKT. For instance, it can distinguish between a learner requiring foundational remediation and one exhibiting high acquisition momentum even when their immediate local performance appears indistinguishable. By integrating these capabilities, gTransformer provides a foundation for truly individualized student-centered instruction, allowing educators to move beyond predicting *if* a student will succeed toward understanding *why* they are struggling.

We evaluate our approach across four primary dimensions. First, we demonstrate parameter recovery accuracy by computing Pearson correlations and generating scatter plot visualizations between the model's predicted parameters and theoretical targets from a fitted BKT model, achieving correlations exceeding 0.7 for both initial mastery and learning rate estimates. Second, we conduct ablation studies to quantify the interpretability cost, showing that theory-guided grounding achieves full interpretability with less than 0.5 percentage points of predictive accuracy loss. Third, we employ a dual evaluation protocol that separately measures both standard supervised predictions and interpretable predictions, demonstrating that the latter outperform classical BKT AUC by averaging 9.0% and up to 19.4%, while remaining highly competitive with non-interpretable DKT baselines. Finally, we present case study visualizations of learning trajectories across distinct learning situations characterized by different combinations of student prior knowledge and learning rates. These analyses show how gTransformer's predictions are context-aware, effectively incorporating the history of interactions in a way that Markovian models like BKT are incapable of. By capturing these behavioral nuances, the model provides longitudinal, individualized diagnostics that support pedagogical decisions such as appropriate placement, pacing, and targeted intervention.

Our research is guided by three primary research questions:

- RQ1: **Theory-Based Interpretability Through Grounded Transformers**
  Can DKT models achieve state-of-the-art predictive performance while providing interpretability grounded in established principles and theories? Specifically, can we design a transformer architecture that produces pedagogically meaningful mastery estimations that are explainable through a causal and interpretable logic, such as BKT?
- RQ2: **Trade-Offs Between Predictive Performance and Interpretability**
  How do the metrics of the supervised, interpretable, and BKT predictions compare? What is the cost of interpretability in terms of AUC? How much predictive gain do the interpretable grounded predictions achieve compared to traditional BKT?
- RQ3: **Practical Value for Student-Centered Personalization**
  Beyond providing interpretable diagnostics, does the high capacity of gTransformer to capture intricate interaction patterns offer advantages over traditional models? Specifically, can these capabilities be leveraged to enhance student-centered personalization relative to population-based models such as BKT?

The remainder of this paper is structured as follows. Section 2 reviews related work, while Section 3 describes the gTransformer model, detailing the construction of embeddings, the grounding mechanism, and the prediction logic. Section 4 presents the experimental setup, followed by a discussion of the empirical results in Section 5. Finally, Section 6 provides the concluding remarks.

2. Related Work

This section provides a review of research related to gTransformer. We first examine the theoretical foundations and limitations of Bayesian Knowledge Tracing, which serves as our reference model for pedagogical grounding. We then trace the evolution of Deep Knowledge Tracing from recurrent neural networks to recent attention-based architectures. Following this, we categorize existing efforts to bridge the interpretability gap in DKT, contrasting post-hoc and ante-hoc methodologies. Finally, our model is mapped to taxonomic dimensions of Informed Machine Learning, establishing the connection with this paradigm.

2.1. Bayesian Knowledge Tracing

For this study, we selected BKT [1] as our reference theoretical framework. It models the learning process as a hidden Markov process governed by four parameters providing a causal narrative of student performance that is intuitive to educators and aligned with cognitive science principles:

- **Initial Mastery**: The probability a student knows a skill prior to practice
- **Learn Rate**: The probability of transitioning from unlearned to learned states after a practice opportunity
- **Guess**: The probability of a correct response despite a lack of mastery
- **Slip**: The probability of an incorrect response despite mastery

Standard BKT suffers from a significant limitation: it typically operates at a population level, estimating a single set of parameters for all students interacting with a given skill. While *Individualized BKT* approaches exist, they are often computationally expensive and struggle with data sparsity [18]. The Representational Grounding used in the gTransformer model offers a new way to address individualization by leveraging the capabilities of Transformers to compress student interaction history into a context vector. This allows the model to infer personalized initial knowledge and learning rates dynamically, transforming population-level BKT insights into high-granularity student diagnostics.

2.2. Deep Knowledge Tracing

The application of deep learning to KT was initiated by the DKT model [8], which framed student performance as a sequence modeling task. While pioneering, DKT was constrained by the inherent limitations of Recurrent Neural Networks (RNNs). These architectures are difficult to train [19] and their sequential nature precludes parallelization within training examples, a bottleneck that becomes critical at longer sequence lengths where memory constraints limit efficiency[9].

The introduction of attention-based architectures, such as Self-Attentive Knowledge Tracing (SAKT) [11] and Separated Self-AttentIve Neural Knowledge Tracing (SAINT) [20], marked a significant shift toward the attention-based Transformers that characterizes modern Large Language Models. They leverage this architecture ability to capture long-range dependencies, allowing for a non-Markovian representation of a student's entire history. More recently, Attentive Knowledge Tracing (AKT) [12] further refined this by incorporating skill difficulty and context-aware attention mechanisms that weigh past interactions based on their temporal distance.

Beyond the prominent recurrent and attention-based paradigms, the research community has explored several other architectural directions to capture the multifaceted nature of student learning. Notable categories include Memory-Augmented models that utilize external storage to explicitly track concept mastery, Graph-Based approaches that model the topological relationships between skills, and specialized variations such as Text-Aware or Forgetting-Aware models that incorporate exercise semantics and temporal decay into the tracing process [2]. While these innovations have further enhanced predictive accuracy, they share the common challenge of balancing model complexity with the pedagogical transparency required for meaningful educational intervention.

2.3. Interpretability in Deep Knowledge Tracing

Interpretability in DKT remains an open challenge despite a variety of attempts, which can be categorized into two primary modalities: *post-hoc* and *ante-hoc*. Post-hoc methods attempt to explain trained black-box models by utilizing either model-specific techniques, such as Layer-Wise Relevance Propagation (LRP) [21], or model-agnostic approaches like Local Interpretable Model-agnostic Explanations (LIME) [22]. However, Rudin [16] argues that relying on post-hoc explanations for high-stakes decision-making is fundamentally flawed. For teachers and domain experts, these methods offer limited utility as they produce explanations tied to technical model structures rather than pedagogically sound principles, necessitating significant technical expertise for meaningful interpretation.

The second category, *ante-hoc* methods, pursues intrinsic interpretability through two primary approaches. The first consists of traditional models that rely on transparent structures, such as BKT or Factor Analysis Models. The second encompasses DKT variants that achieve interpretability by incorporating additional modules or utilizing side information [10]. Among these, attention mechanisms have emerged as the most prominent architectural extensions. They have been integrated into a wide range of models, from the pioneering SAKT [11] to more recent architectures like AKT [12], which achieves high predictive performance by utilizing multi-scale attention to capture interaction patterns across varying temporal resolutions. However, relying solely on attention weights presents similar practical obstacles to those of post-hoc methods for educational practitioners, as these mechanisms do not provide an explicit interpretation grounded in pedagogically sound principles.

The second approach to ante-hoc interpretability involves the integration of side information, with Item Response Theory (IRT) [6] serving as the most prominent foundation. IRT estimates the probability of a correct response based on a student's latent ability and the item's difficulty. This theory has been coupled with DKT models such as Deep-IRT [13], which utilizes dynamic key-value memory networks to capture learners' trajectories and uses inferred abilities and difficulties to predict performance. Similarly, TC-MIRT [14] embeds multidimensional IRT parameters to predict student states and generate interpretable parameters. Other relevant models include KIKT [23], which harnesses IRT to simulate student performance; LANA [24], which adopts an IRT variant for ability clustering; and QIKT [15], which features question-centric representations integrated with an interpretable IRT layer. Beyond IRT, researchers have leveraged diverse principles, such as constructive learning [25,26], learning and forgetting curves [27], finite state automata (FSAs) [28], classical test theory (CTT) [29], and monotonicity constraints [30].

In contrast to these models, whose diagnostic utility is often restricted by the specificity and scope of the single underlying theory serving as side information [10], gTransformer offers a more flexible framework. It can be grounded in any established theoretical reference, effectively decoupling the expressiveness of deep learning from the constraints of specific pedagogical models.

2.4. Informed Machine Learning

The integration of domain knowledge into deep learning has been broadly explored through conceptual frameworks such as Theory-Guided Data Science (TGDS) [31], Physics-Informed Machine Learning (PIML) [32], and Informed Machine Learning (IML) [17].

TGDS emerged as a paradigm to address the limitations of purely data-driven "black-box" models, which may produce results inconsistent with established domain knowledge [31]. By explicitly requiring scientific consistency, TGDS integrates theoretical insights into the learning process to ensure that models remain scientifically plausible. This integration is typically achieved via theory-guided learning where domain-specific invariants serve as regularization terms—or through the design of neural architectures that respect the structural constraints of the problem [31,33]. A prominent implementation of this paradigm is Physics-Informed Neural Networks (PINNs) [32,34], which embed differential equations directly into the loss function, forcing the model to satisfy theoretical constraints while learning from empirical data.

A widely adopted technique to enforce this consistency involves incorporating domain-specific constraints into the loss function as follows [31,35]:

$$\mathcal{L} = \mathcal{L}_{TRN}(Y_{true}, Y_{pred}) + \gamma \mathcal{L}_{THEORY}(Y_{pred}), \tag{1}$$

where $\mathcal{L}_{TRN}$ measures the supervised error between true labels Y_{true} and predicted labels Y_{pred}, while $\mathcal{L}_{THEORY}$ penalizes violations of theoretical constraints, weighted by the hyperparameter γ.

Von Rueden et al. [17] formalized these efforts into an IML taxonomy, which categorizes knowledge integration by its source, representation, and the stage of the machine learning pipeline where it is incorporated. Knowledge can be represented as algebraic equations, logical rules, or probabilistic priors, and integrated at various stages: within the training data, through the design of tailored architectures, or by influencing the model's final output.

We operationalized these taxonomic dimensions in gTransformer by integrating theoretical priors from an interpretable BKT model into the input layer and employing a multi-objective loss function to anchor the model's predictions to BKT parameters. However, while most IML approaches focus on enhancing predictive performance [17], gTransformer aims to achieve theory-based interpretability by grounding estimations in sound pedagogical concepts, as detailed in Section 3.1.

3. The gTransformer Model

This section details the design and implementation of gTransformer, a neuro-symbolic architecture that bridges the gap between deep learning expressiveness and theoretical alignment by grounding the Transformer's latent representations in concepts established by a reference model. We begin by mapping its components to the taxonomic dimensions of IML, clarifying how theoretical priors and empirical data are processed using specialized inductive biases and multi-objective optimization. We then provide a comprehensive description of the model's architecture, detailing its functional layers: from embeddings enriched with prior knowledge to the core attention-based encoder-decoder engine. We explain the projection mechanism enabling the estimation of individualized, pedagogically meaningful parameters, which are fed into an output head implementing BKT logic to produce interpretable predictions. Finally, we detail the loss functions that anchor the model's predictions to BKT parameters.

3.1. Dimensions of Informed Machine Learning in gTransformer

Figure 1 illustrates the information flow through gTransformer, mapped to the taxonomic dimensions of IML as defined by Von Rueden et al. [17].

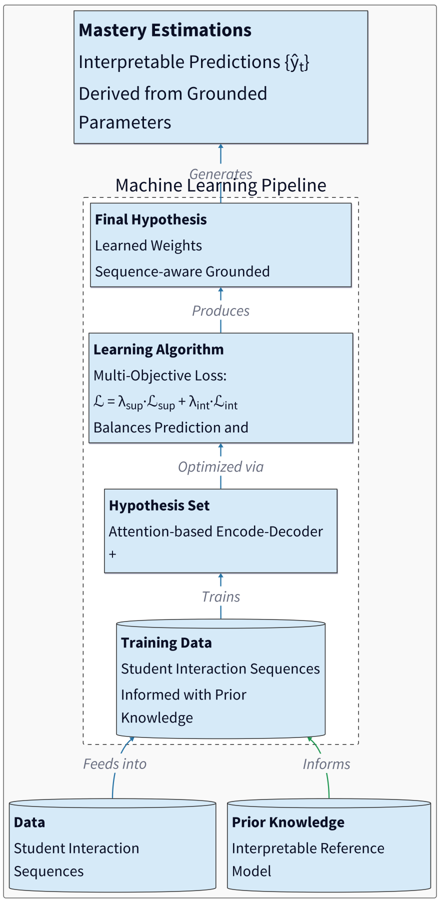

Figure 1. Dimensions of Informed Machine Learning in gTransformer.

Within the context of our architecture, these dimensions are operationalized as follows:

- **Prior Knowledge**: The source of theoretical information comprises simulation results from a reference model, typically expressed as parameter values that represent foundational pedagogical constructs.
- **Training Data**: Empirical student interaction histories are augmented with theoretical priors to guide the learning process toward pedagogically valid states.
- **Hypothesis Set**: An attention-based Transformer architecture is employed to provide the inductive biases necessary for grounding internal latent representations within the conceptual framework of the reference model.
- **Learning Algorithm**: The model is optimized via a multi-objective loss function formulated to balance the trade-off between predictive performance and pedagogical interpretability.
- **Final Hypothesis**: A set of learned weights that drive latent representations from which parameter values aligned with the reference model can be extracted.

3.2. Architecture

The gTransformer architecture, illustrated in Figure 2, implements a theory-guided deep learning framework that integrates BKT concepts throughout the processing pipeline. The architecture is organized into six functional layers, which are detailed in the following subsections. Specifically, we: (i) begin with input data comprising student interaction sequences along with prior knowledge from the Bayesian reference model (Section 3.2.1); (ii) proceed to difficulty-aware embeddings that encode skill-specific variations (Section 3.2.2); (iii) describe the attention-based encoder-decoder mechanism used to construct contextualized latent representations (Section 3.2.3); (iv) explain how grounded parameters are derived through additive composition in logit space, combining theoretical bases with contextual projections onto semantic axes (Section 3.2.4); (v) present the three parallel output heads intended to generate supervised predictions, interpretable predictions, and representation probes (Section 3.2.5); and (vi) conclude with the multi-objective loss function that orchestrates the training process to achieve both predictive performance and theory-based interpretability (Section 3.2.6).

3.2.1. Input Components

The data flow begins with the ingestion of student interaction sequences enriched with population-level parameters from a pre-fit BKT reference model. Student interactions are represented as temporal sequences of tuples (q_t, c_t, r_t), where q_t denotes the question identifier, c_t represents the associated knowledge component, concept or skill, and $r_t \in \{0, 1\}$ indicates the binary correctness of the student's response at timestep t.

These empirical interaction traces are augmented with theoretical priors extracted from the BKT reference model. For each knowledge component c, we obtain population-level estimates of four fundamental BKT parameters: prior knowledge $P(L_0)_c$ (the probability that a randomly selected student has already mastered concept c before any practice), learning rate $P(T)_c$ (the probability of transitioning from non-mastery to mastery after a single learning opportunity), guess probability $P(G)_c$ (the likelihood of answering correctly despite not having mastered the concept), and slip probability $P(S)_c$ (the likelihood of answering incorrectly despite having mastered the concept). These parameters serve as the theoretical anchor that grounds the model's internal representations, ensuring that learned latent features remain pedagogically interpretable throughout the deep learning pipeline.

3.2.2. Prior-Enriched Embeddings

Rather than treating questions and responses as discrete atomic identifiers, gTransformer decomposes them into structured embedding vectors that capture both semantic identity and task-specific variation. This decomposition follows a Rasch-inspired addi-

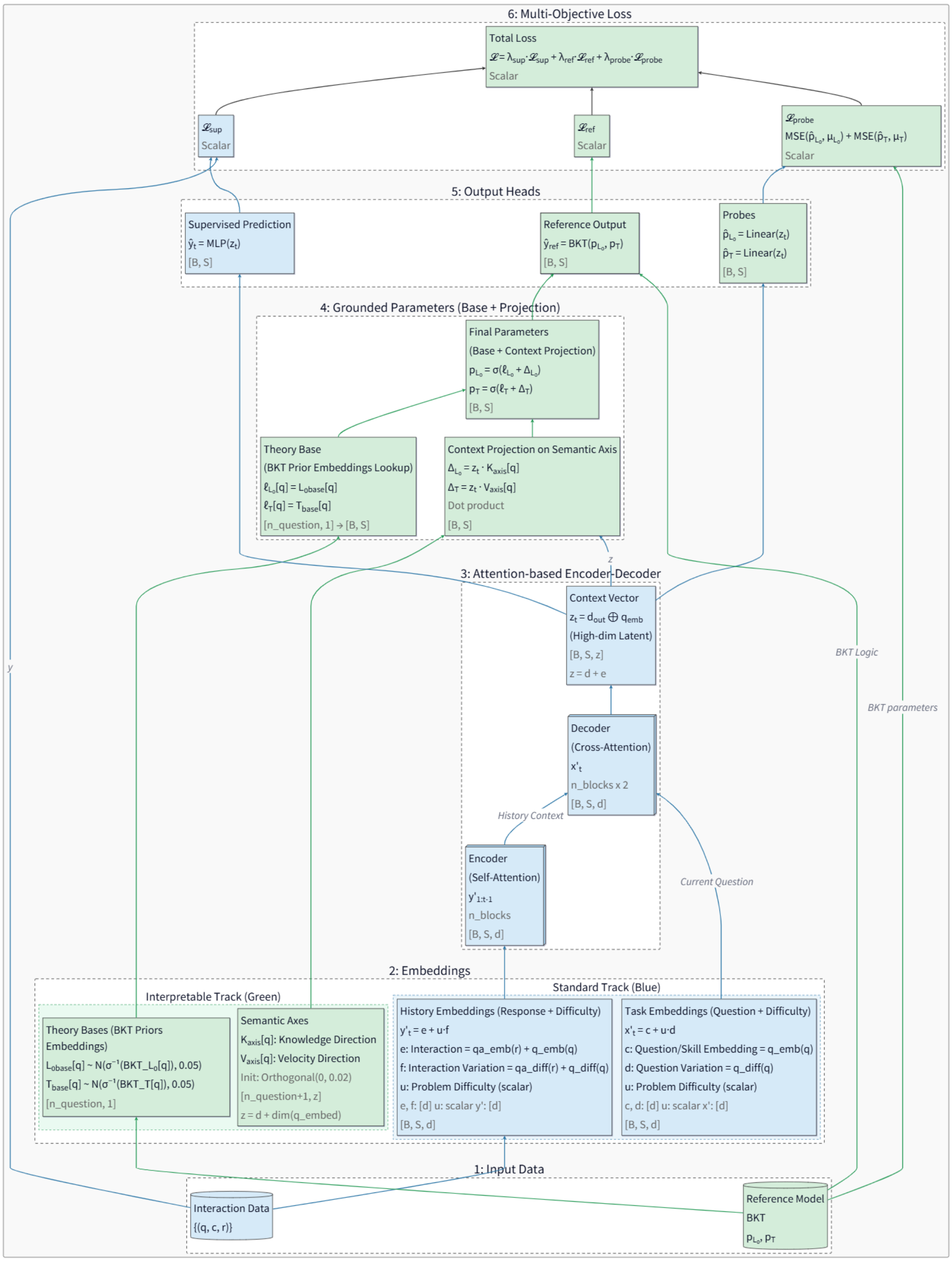

Figure 2. Functional layers of the gTransformer Architecture (blue components denote standard Transformer elements; green components denote gTransformer-specific theoretical grounding): (1) *Input Components*: student interaction data enriched with BKT reference parameters; (2) *Difficulty-Aware Embeddings*: task-specific embeddings (x') composed of concept bases and difficulty variations, and interaction history embeddings (y') encoding prior responses; (3) *Attention-based Encoder-Decoder*: a stacked Transformer architecture with n encoder blocks (self-attention on interaction history) and $2n$ decoder blocks (alternating self-attention on current questions and cross-attention with encoded history), producing a high-dimensional context vector z_t by concatenating the final decoder output d_{out} with question embeddings; (4) *Grounded Parameters*: additive composition of BKT-initialized theoretical bases (ℓ_{L_0}, ℓ_T) and contextual projections (Δ_{L_0}, Δ_T) computed via dot products of z_t with learnable semantic axes (K_{axis}, V_{axis}), yielding pedagogically-grounded parameters (p_{L_0}, p_T); (5) *Output Heads*: supervised prediction ($\hat{y}_{sup}$) from an MLP applied to z_t, interpretable prediction ($\hat{y}_{ref}$) generated by feeding enriched parameters (projected from z_t) through BKT logic to preserve causal explainability (Reference Output), and linear probes ($\hat{p}_{L_0}, \hat{p}_T$) for internal representation analysis; finally (6) *Multi-Objective Loss*: weighted combination of supervised loss ($\mathcal{L}_{sup}$), reference output loss ($\mathcal{L}_{ref}$), and probing loss ($\mathcal{L}_{probe}$) for aligning neural representations with pedagogical theory.

tive composition principle, where each embedding is constructed as the sum of a base representation and a difficulty-modulated perturbation.

For question embeddings, we define:

$$x'_t = c_q + u_q \cdot d_c \tag{2}$$

where $c_q \in \mathbb{R}^d$ represents the invariant concept base embedding (encoding the semantic identity of the knowledge component), $u_q \in \mathbb{R}$ is a scalar difficulty parameter specific to question q, and $d_c \in \mathbb{R}^d$ is the concept-specific difficulty variation axis. This formulation allows multiple questions targeting the same concept to share a common semantic core (c_q) while differing in their position along a learned difficulty direction (d_c), with the magnitude of displacement controlled by the per-question difficulty scalar (u_q).

Similarly, for interaction history embeddings (encoding both the question and the student's response), we employ:

$$y'_t = e_{c,r} + u_q \cdot f_{c,r} \tag{3}$$

where $e_{c,r} \in \mathbb{R}^d$ is the interaction base embedding (jointly encoding the concept c and response outcome r), and $f_{c,r} \in \mathbb{R}^d$ is the interaction-specific difficulty variation axis. These embeddings serve as the primary inputs to the Transformer's attention mechanism, providing multi-dimensional representations that enable attention-driven discovery of relevant behavioral patterns across the student's learning history.

3.2.3. Attention-Based Encoder-Decoder

The core processing engine of gTransformer is a stacked Transformer architecture composed of n encoder blocks followed by $2n$ decoder blocks. The encoder processes the student's longitudinal interaction history through n sequential self-attention layers, transforming the sequence of interaction embeddings $\{y'_1, y'_2, \ldots, y'_{t-1}\}$ into a tensor of contextualized latent representations $\mathbf{Y}_{\text{enc}} \in \mathbb{R}^{(t-1)\times d}$. Each encoder block applies multi-head self-attention with monotonic causal masking to prevent information leakage from future timesteps, followed by position-wise feed-forward networks and residual connections with layer normalization.

The decoder architecture employs a distinctive alternating pattern of $2n$ sequential layers. Odd-numbered decoder layers (layers $1, 3, 5, \ldots, 2n-1$) perform self-attention over the current question embeddings x'_t, allowing the model to examine multiple aspects of the current task without incorporating response history. Even-numbered decoder layers (layers $2, 4, 6, \ldots, 2n$) perform cross-attention between the current question representation and the encoder's output $\mathbf{Y}_{\text{enc}}$, retrieving relevant knowledge from the student's encoded learning trajectory. Critically, only the even-numbered cross-attention layers include position-wise feed-forward networks, while odd-numbered self-attention layers omit them. This asymmetric design creates n complete encoder-decoder units, where each unit consists of a self-attention layer (to process current task features) paired with a cross-attention layer (to integrate historical context).

The final decoder layer produces an output tensor $\mathbf{d}_{\text{out}} \in \mathbb{R}^{t\times d}$, which is concatenated with the question embedding to form the high-dimensional context vector:

$$z_t = [\mathbf{d}_{\text{out},t}; x'_t] \in \mathbb{R}^z \tag{4}$$

where $z = d + e$ represents the combined dimensionality of the decoder output and question embedding spaces. This context vector serves as a comprehensive latent encoding of the student's current cognitive state, synthesizing information from both the observed interaction history and the current task characteristics.

3.2.4. Grounded Parameters

To ensure pedagogical interpretability, gTransformer estimates BKT parameters through an additive composition mechanism operating in logit space. For each knowledge component c, we define theoretical base parameters $\ell_{L_0,c}$ and $\ell_{T,c}$, initialized as the logit-transformed population-level BKT priors:

$$\ell_{L_0,c} = \text{logit}(P(L_0)_c), \quad \ell_{T,c} = \text{logit}(P(T)_c) \tag{5}$$

These bases are implemented as learnable scalar embeddings with Gaussian initialization centered at the theoretical values ($\mathcal{N}(\mu = \text{logit}(P), \sigma = 0.05)$), ensuring that the model begins training from pedagogically valid priors while allowing fine-tuning through gradient descent.

Contextual adjustments are computed via dot products between the latent context vector z_t and learnable semantic axes. For each concept c, we define a knowledge axis $\mathbf{K}_c \in \mathbb{R}^z$ and a learning velocity axis $\mathbf{V}_c \in \mathbb{R}^z$. The contextual projections, representing the student-specific Knowledge Gap (k_s) and Learning Velocity (v_s), are calculated as:

$$k_{s,t} = z_t \cdot \mathbf{K}_c, \quad v_{s,t} = z_t \cdot \mathbf{V}_c \tag{6}$$

These dot products project the high-dimensional latent state onto interpretable semantic dimensions, quantifying how much the student's observed behavior deviates from the population average along the dimensions of prior mastery and learning rate.

The final grounded parameters are obtained through additive composition in logit space, followed by sigmoid activation to ensure valid probability bounds:

$$p_{L_0,t} = \sigma(\ell_{L_0,c} + k_{s,t}), \quad p_{T,t} = \sigma(\ell_{T,c} + v_{s,t}) \tag{7}$$

This design ensures that the model's parameter estimates remain structurally anchored to BKT theory—the base terms provide pedagogically sound initialization, while contextual projections enable individualized adjustments driven by observed student behavior.

3.2.5. Output Heads

The gTransformer model generates predictions through three parallel output heads, each serving a distinct functional role in the architecture's dual objectives of predictive accuracy and pedagogical interpretability.

Supervised Prediction Head: A multi-layer perceptron (MLP) directly maps the context vector z_t to a correctness prediction $\hat{y}_t = \text{MLP}(z_t)$. This head is optimized purely for predictive accuracy via binary cross-entropy loss, leveraging the full expressive capacity of the Transformer's learned representations without theoretical constraints.

Interpretable Output Head: The grounded BKT parameters $(p_{L_0,t}, p_{T,t})$ are passed through a differentiable implementation of BKT logic, which performs a retrospective belief update walk over the student's interaction history. At each timestep, the BKT logic computes the probability of correctness $\hat{y}_{\text{ref},t}$ using the standard BKT update equations with the model's estimated mastery probabilities and the fixed population-level guess and slip parameters. This head ensures that the grounded parameters retain their intended pedagogical semantics by requiring them to produce valid predictions when subjected to BKT's causal reasoning process.

Linear Probe Heads: Two dedicated linear layers extract BKT parameter estimates directly from the context vector: $\hat{p}_{L_0,t} = \sigma(\mathbf{W}_{L_0} z_t)$ and $\hat{p}_{T,t} = \sigma(\mathbf{W}_T z_t)$. Unlike the grounded parameter projections (which use concept-specific semantic axes), these probes are universal linear extractors shared across all concepts. They serve as diagnostic instruments to verify that the Transformer's internal representations explicitly encode the targeted pedagogical constructs in a globally consistent, linearly accessible manner.

3.2.6. Multi-Objective Loss

The training objective balances predictive performance with pedagogical grounding through a weighted combination of three loss terms:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{sup}} \mathcal{L}_{\text{sup}} + \lambda_{\text{ref}} \mathcal{L}_{\text{ref}} + \lambda_{\text{probe}} \mathcal{L}_{\text{probe}} \tag{8}$$

The supervised loss $\mathcal{L}_{\text{sup}}$ measures the binary cross-entropy between the MLP's direct predictions $\hat{y}_t$ and the ground truth responses r_t, driving the model to maximize predictive accuracy:

$$\mathcal{L}_{\text{sup}} = -\frac{1}{N} \sum_{t=1}^{N} [r_t \log(\hat{y}_t) + (1 - r_t) \log(1 - \hat{y}_t)] \tag{9}$$

The reference output loss $\mathcal{L}_{\text{ref}}$ applies the same binary cross-entropy metric to the BKT logic head's predictions $\hat{y}_{\text{ref},t}$, ensuring that the grounded parameters $(p_{L_0,t}, p_{T,t})$ produce pedagogically meaningful predictions when constrained to operate through BKT's causal inference mechanism:

$$\mathcal{L}_{\text{ref}} = -\frac{1}{N} \sum_{t=1}^{N} [r_t \log(\hat{y}_{\text{ref},t}) + (1 - r_t) \log(1 - \hat{y}_{\text{ref},t})] \tag{10}$$

The probing loss $\mathcal{L}_{\text{probe}}$ enforces global alignment between the Transformer's latent representations and BKT constructs by minimizing the mean squared error between linear probe predictions and population-level BKT targets:

$$\mathcal{L}_{\text{probe}} = \sum_{k \in \{L_0, T\}} \text{MSE}(\hat{p}_{k,t}, P(k)_c) \tag{11}$$

This loss implements an "active grounding" mechanism, forcing the context vector z_t to organize its internal structure such that pedagogical parameters can be extracted via simple linear transformations. By constraining the latent space globally rather than merely regularizing output parameters locally, the probing loss ensures that interpretability is structurally embedded throughout the model's representations.

The default configuration uses weight coefficients $\lambda_{\text{sup}} = 1.0$, $\lambda_{\text{ref}} = 0.5$, and $\lambda_{\text{probe}} = 1.0$, balancing supervised performance with theoretical grounding while prioritizing global latent space alignment through active probing.

4. Experimental Setup

This section describes the experimental setup used to evaluate gTransformer. We first detail the architecture configuration and training protocol, including the set of hyperparameters and the cross-validation procedure. We then describe the five educational datasets used for evaluation, which are among the most commonly used in DKT benchmarks.

4.1. Architecture and Training Setup

The gTransformer model was implemented in `PyTorch` using the `pyKT` library [36] for standardized data preprocessing, dataset splitting, and baseline models. Baseline models were configured using hyperparameter tuning results from the official pyKT repository.

As detailed in Table 1, the gTransformer architecture was configured with an embedding dimension $d_{model} = 64$, four attention heads, four encoder blocks, and eight decoder blocks, with a feed-forward dimension $d_{ff} = 256$.

Training was conducted using standard five-fold cross-validation with an 80/20 train/test split. We employed the Adam optimizer with a learning rate of 10^{-4}, a batch size of 64, and a dropout rate of 0.2 to mitigate overfitting. Training was limited to a maximum of 200 epochs, with an early stopping patience of 10 epochs. Predictive performance was evaluated using the Area Under the Curve (AUC) following the question-level average late-fusion protocol recommended in [36], while interpretability was assessed via the metrics defined in Section 5.2.

Parameter	**AS2009**	**AS2015**	**AL2005**	**BDG2006**	**NIPS34**
Architecture Configuration					
Embedding dimension (d_{model})	64	64	64	64	64
Attention heads	4	4	4	4	4
Encoder blocks	4	4	4	4	4
Decoder blocks	8	8	8	8	8
Feed-forward dimension (d_{ff})	256	256	256	256	256
Training Configuration					
Optimizer	Adam	Adam	Adam	Adam	Adam
Learning rate	2×10^{-4}	10^{-4}	10^{-4}	10^{-4}	2×10^{-4}
Batch size	64	64	64	64	64
Dropout rate	0.1	0.1	0.1	0.1	0.1
Epochs	200	200	200	200	200

Table 1. Parameter values used to train gTransformer across all datasets

4.2. Datasets

We selected five educational datasets and six models that are between the most frequently mentioned DKT datasets and baselines [36] as representative DKT datasets and methods to evaluate our model and compare it.

- ASSISTments 2009 (AS2009): A benchmark dataset from the ASSISTments platform (2009-2010), representing a standard for KT evaluation [37].
- ASSISTments 2015 (AS2015): A large-scale release from the same platform with a high volume of student records [37].
- Algebra 2005 (AL2005): From the KDD Cup 2010 EDM Challenge, containing 13-14 year old students' responses to Algebra questions with detailed step-level student responses [38].
- Bridge to Algebra 2006 (Bridge2006): A dataset from the KDD Cup 2010 EDM Challenge focused on the Carnegie Learning Bridge to Algebra system [38].
- NeurIPS 2020 Education Challenge (NIPS34): A diagnostic dataset from the Eedi platform containing multiple-choice math questions [39].

5. Results and Discussion

5.1. Predictive Performance

To evaluate the predictive performance of gTransformer, we utilized the Area Under the Curve (AUC) across five benchmark datasets as described in Section 4. The experimental results are summarized in Table 2.

When comparing these results with baseline architectures, we observe that gTransformer achieves highly competitive performance across all benchmarks. For this comparison, we utilize a version of gTransformer with interpretability mechanisms ablated,

Model	AS2009	AS2015	AL2005	BDG2006	NIPS34
AKT	0.7825	0.7081	0.8306	0.8208	0.8033
ATKT	0.7715	0.7810	0.7995	0.7889	0.7665
DKT	0.7528	0.7031	0.8149	0.8015	0.7689
DKVMN	0.7440	0.7013	0.8054	0.7983	0.7673
SAKT	0.7263	0.6947	0.7880	0.7740	0.7517
SAINT	0.6921	0.6836	0.7775	0.7781	0.7873
gTransformer	0.7831	0.7078	0.8240	0.8148	0.8006

Table 2. AUC (question-level, average late fusion) across the evaluated datasets.

functioning as a pure neural model to establish its maximum representational capacity. On AS2009, this configuration achieves 0.7831, outperforming all other models. On all other datasets, it achieves second place after AKT.

5.2. Theory-Based Interpretability Through Grounded Transformers (RQ1)

We will use the following hypotheses to validate the RQ1 research question posed in section 1:

- **H1.1: Structural Encoding**
  Latent representations in the gTransformer model are structurally organized around BKT constructs (initial mastery P_{L0} and learning rate P_T) as the dominant organizing principle.
- **H1.2: Semantic Alignment**
  The extracted parameters are semantically aligned with meaningful BKT priors.
- **H1.3: Functional Alignment**
  The interpretable predictions derived from extracted parameters can be used with quantified confidence, enabling educators to identify when theory-grounded explanations are trustworthy versus when additional validation is recommended.

The objective of this triple validation is to demonstrate that grounding constraints actively shape the model's internal structure, not merely correlate with outputs, producing interpretable predictions. It addresses a fundamental challenge in educational AI: whether neural models can internalize pedagogical theory rather than learning arbitrary features that happen to predict well, and whether such interpretability can be achieved without sacrificing the representational power that makes deep learning effective.

5.2.1. Structural Alignment via Diagnostic Probing (H1.1)

To verify whether gTransformer's latent representations genuinely internalize pedagogical concepts, we employ *diagnostic probing with control tasks* [40,41]. This approach trains linear Ridge regression models ($\alpha = 1.0$) to act as diagnostic probes, checking if the original BKT theoretical parameters (p_{L_0} and p_T) can be extracted from the final hidden states z_t produced by gTransformer.

Methodology: We measure *fidelity* as the quality of the parameter recovery from the latent representations, measured via R^2 and Pearson correlation r using an 80/20 train-test split. To distinguish genuine structural encoding from spurious correlations, *selectivity* is calculated to compare probe performance on true BKT targets against a control task with randomly shuffled target values. This distinction verifies whether probes exploit genuine BKT structure rather than arbitrary patterns.

The rationale is that high selectivity (> 0.5) proves BKT constructs are the primary organizing axis rather than incidental patterns—if probes perform equally well on shuffled targets, the representations would contain no genuine structural encoding.

- **Fidelity**: The R^2 (coefficient of determination) is calculated on the test set, representing the proportion of the variance in the theoretical parameters that is predictable from the gTransformer's latent space:

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

  where y_i is the actual BKT theoretical value, $\hat{y}_i$ is the value predicted by the linear probe from the hidden state z_t, and $\bar{y}$ is the mean of the theoretical values.
- **Selectivity**: Quantifies the structural specificity by comparing true performance against a shuffled control:

$$\Delta R^2 = R^2_{\text{true}} - R^2_{\text{control}}$$

  where a negative R^2_{control} signifies that the probe performs worse than a mean-based baseline on shuffled data, further confirming the structural specificity of the true representations.

Results: Table 3 summarizes the diagnostic probing results on AS2009. The recorded values exceed the $R^2 = 0.5$ threshold, indicating strong structural encoding:

- **Initial Mastery (L_0)**: Achieves fidelity $R^2 = 0.487$ with Pearson $r = 0.698$, indicating strong linear recoverability. The selectivity $\Delta R^2 = 0.507$ exceeds the threshold of 0.5, proving L_0 is a dominant organizing principle in the latent space rather than an incidental pattern. The control task achieves negative $R^2 = -0.020$, confirming probes cannot recover shuffled targets.
- **Learning Rate (T)**: Shows even stronger structural encoding with fidelity $R^2 = 0.566$ and Pearson $r = 0.753$. The selectivity reaches $\Delta R^2 = 0.607$, the highest among both constructs, with control $R^2 = -0.041$. This superior selectivity suggests learning rate trajectories impose more distinctive constraints on latent geometry than initial knowledge.

Interpretation: The high selectivity (> 0.5) for both parameters validates H1.1 (Structural Encoding) hypothesis: BKT constructs are preserved through all Transformer layers and remain as primary organizing axes in final representations used for prediction. The negative control R^2 values confirm probes exploit genuine structure rather than overfitting noise, demonstrating that the model's latent organization is fundamentally shaped by pedagogical theory rather than arbitrary statistical patterns.

Figures 3 and 4 visualize the recovery quality through parity plots comparing BKT theoretical priors (x-axis) against probe-extracted parameters (y-axis). Each point represents a binned aggregate of test interactions, with size encoding sample density. The strong adherence to the diagonal (red dashed line) confirms linear recoverability. The T plot exhibits more linear fidelity across the full range, consistent with its higher R^2 and selectivity scores.

5.2.2. Semantic Grounding and Alignment Preservation (H1.2)

Hypothesis H1.2 examines whether grounded parameters $\{p_{L_0,t}, p_{T,t}\}$ retain their pedagogical semantics after neural processing. We want to check that models do not repurpose theoretical constructs for black-box optimization. Unlike H1.1, which assesses latent linear decodability, H1.2 validates monotonic relationship preservation in the final projected parameters. Because gTransformer refines population-level priors into student-

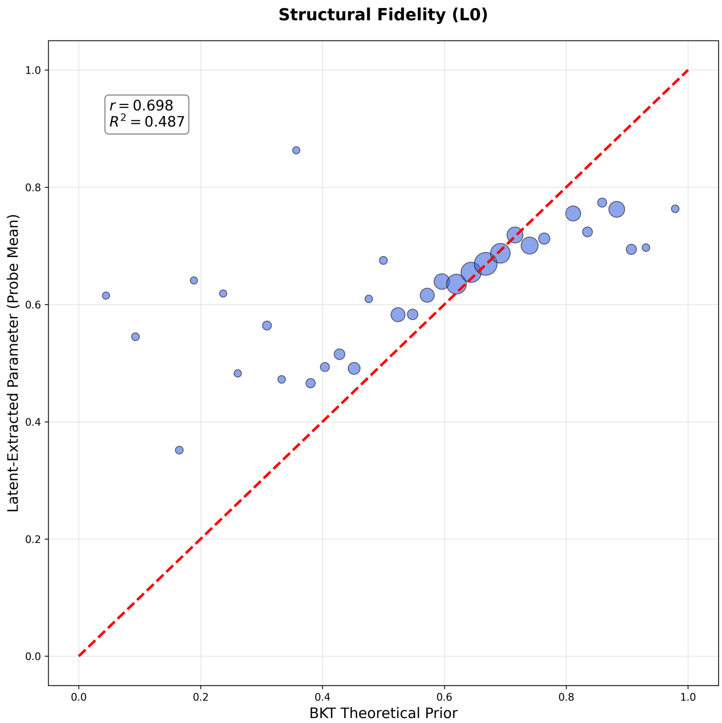

Figure 3. Structural Fidelity for Initial Mastery P_{L_0} (H1.1). Parity plot showing strong linear recoverability ($r = 0.698$, $R^2 = 0.487$) between BKT theoretical priors and probe-extracted parameters. Each bubble represents a binned aggregate of test interactions, with size encoding sample density. The tight adherence to the diagonal (red dashed) validates that latent representations preserve P_{L_0} structure across the full parameter range.

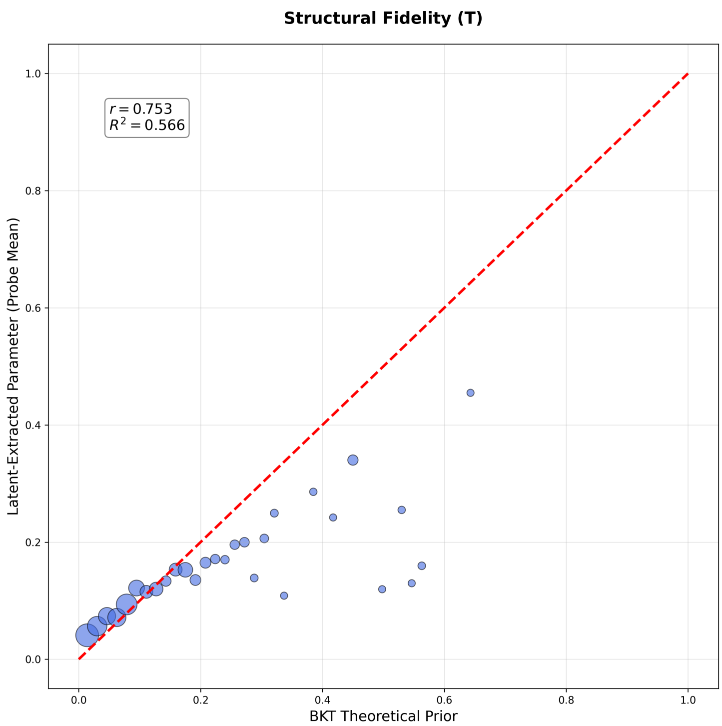

Figure 4. Structural Fidelity for Learning Rate P_T (H1.1). Parity plot demonstrating robust linear recoverability ($r = 0.753$, $R^2 = 0.566$) with superior fidelity compared to P_{L_0}. The consistently tight clustering around the diagonal across all P_T values indicates the model encodes learning rate dynamics with high precision.

Construct	Fidelity	Pearson (r)	Control	Selectivity (Δ)
Initial Mastery (L_0)	0.487	0.698	−0.020	**0.507**
Learning Rate (T)	0.566	0.753	−0.041	**0.607**

Table 3. Diagnostic Probing results for AS2009 (H1.1). Fidelity measures probe accuracy on true BKT targets; selectivity quantifies performance advantage over randomly shuffled controls. High selectivity (> 0.5) proves BKT constructs are dominant organizing principles in latent representations, while negative control R^2 confirms probes cannot recover arbitrary patterns.

specific estimates, we prioritize maintaining pedagogical ordering over exact numerical reproduction.

We use Spearman's rank correlation (ρ) to measure this alignment, as it assesses monotonic consistency without assuming linearity (thresholds: $\rho \geq 0.6$ strong; $0.4 \leq \rho < 0.6$ moderate).

Results: Table 4 summarizes the semantic alignment results on AS2009. The grounded parameters achieve moderate-to-strong correlations with theoretical priors:

- **Initial Mastery (P_{L_0})**: Achieves Spearman $\rho = 0.427$ (moderate), demonstrating rank-order preservation despite neural transformation. The lower correlation compared to probe parameters ($\rho = 0.736$ in H1.1) confirms genuine student-specific individualization rather than trivial reproduction of population priors.
- **Learning Rate (P_T)**: Shows stronger alignment with Spearman $\rho = 0.604$ (moderate-to-strong), approaching the 0.6 threshold for strong alignment. This superior preservation suggests the model reliably internalizes theoretical principles of practice effects. The comparison with probe parameters ($\rho = 0.760$ in H1.1) again confirms meaningful contextual refinement.

Interpretation: The moderate-to-strong Spearman correlations validate H1.2 (Semantic Alignment) hypothesis: grounded parameters maintain pedagogical ordering after neural transformation while enabling individualization. The gap between grounded and probe correlations (0.309 for L_0, 0.156 for T) quantifies the degree of context-aware refinement—the model neither abandons theoretical priors ($\rho \approx 0$) nor mechanically reproduces them ($\rho \approx 1$), achieving the desired balance.

Figures 5 and 6 visualize the alignment quality through parity plots comparing BKT theoretical priors (x-axis) against grounded parameters (y-axis). Each point represents a binned aggregate of test interactions, with size encoding sample density. High-density regions (large bubbles) cluster near the diagonal (red dashed line), confirming rank-order preservation where data is abundant while allowing adaptive refinement in sparse regions.

Parameter	Spearman ρ (H1.2)	Probe ρ (H1.1)	Gap (Δ)	Interpretation
Initial Mastery (P_{L_0})	0.427	0.736	0.309	Moderate
Learning Rate (P_T)	0.604	0.760	0.156	Moderate-to-Strong

Table 4. Semantic Alignment results for AS2009 (H1.2). Spearman ρ measures monotonic relationship preservation between grounded parameters and BKT theoretical priors. Comparison with probe-based correlations from H1.1 quantifies the degree of context-aware individualization (Gap Δ). Moderate-to-strong correlations confirm pedagogical semantics are preserved through neural transformation while enabling student-specific refinement.

These results demonstrate a successful balance between theoretical grounding and adaptive individualization. Parity plots show that high-density regions cluster near the theoretical diagonal, indicating that semantic integrity is preserved even during student-specific refinement. These findings support H1.2, confirming that grounded parameters maintain moderate-to-strong monotonic alignment with pedagogical theory while capturing individual behavioral heterogeneity.

5.2.3. Functional Alignment and Prediction Confidence (H1.3)

For the third hypothesis H1.3 (Functional Alignment), we compare two prediction paths: interpretable predictions $\hat{y}_{ref}$ generated by feeding enriched parameters (projected from context vectors $\mathbf{z}_t$) through BKT logic, and supervised predictions $\hat{y}_{sup}$ obtained directly from an MLP applied to $\mathbf{z}_t$. While $\hat{y}_{sup}$ uses the full representational capacity of the neural network for maximum predictive accuracy, $\hat{y}_{ref}$ constrains predictions through pedagogically interpretable BKT parameters, enabling causal explanations at the cost of some predictive power. We assess the *confidence* with which the interpretable reference path can functionally replace the supervised path across different student-skill contexts.

To quantify prediction trustworthiness, we use a *composite confidence metric* that evaluates three critical dimensions:

$$\text{Confidence} = 0.4 \times C_{\text{calibrated}} + 0.3 \times C_{\text{directional}} + 0.3 \times C_{\text{percentile}} \tag{12}$$

where:

- $C_{\text{calibrated}} = \exp(-2|\hat{y}_{ref} - \hat{y}_{sup}|)$ measures numerical closeness with exponential decay (e.g., a 0.1 disagreement yields 0.90 confidence while a 0.5 disagreement yields only 0.14).
- $C_{\text{directional}} = \begin{cases} 1 & \text{if } (\hat{y}_{ref} \geq 0.5 \text{ and } \hat{y}_{sup} \geq 0.5) \text{ or } (\hat{y}_{ref} < 0.5 \text{ and } \hat{y}_{sup} < 0.5) \\ 0 & \text{otherwise} \end{cases}$ ensures binary decision agreement for intervention purposes (both predictions must agree on pass/fail).
- $C_{\text{percentile}} = \frac{100 - \text{percentile\_rank}(|\hat{y}_{ref} - \hat{y}_{sup}|)}{100}$ ranks the current disagreement against all disagreements in the dataset (e.g., if a 0.15 disagreement falls at the 70th percentile, then $C_{\text{percentile}} = 0.30$, indicating this disagreement is larger than 70% of cases).

The three-component design addresses distinct trust requirements: simple numerical proximity can be misleading when predictions fall on opposite sides of the 0.5 decision threshold (e.g., $\hat{y}_{ref} = 0.49$ vs $\hat{y}_{sup} = 0.51$ differ by only 0.02 but yield opposite interventions), while context-free metrics cannot account for skill-specific prediction difficulty—certain skills consistently exhibit higher disagreement between $\hat{y}_{ref}$ and $\hat{y}_{sup}$ across all students, making a given disagreement magnitude acceptable in such cases but problematic for skills where both predictors typically align closely. Using the more accurate supervised predictions $\hat{y}_{sup}$ as a trust anchor, this metric identifies student-skill pairs where interpretable predictions are reliable versus those requiring human review.

Across AS2009 test interactions, we observe heterogeneous confidence patterns (Figure 7), with 89.2% of student-skill pairs achieving medium or high confidence, while only 10.8% exhibit low confidence. This shows that interpretable predictions provide actionable diagnostic value with quantified uncertainty bounds, enabling educators to leverage theory-grounded explanations while recognizing contexts requiring additional validation.

5.3. Accuracy–Interpretability Trade-Off (RQ2)

To quantify the performance trade-off between predictive accuracy and interpretability, we compare three prediction sources across five educational datasets using a single grounded transformer architecture (ablation=none, $d_{\text{model}} = 64$, 4 blocks, 4 attention heads). The model produces two types of predictions: (1) supervised neural predictions (p_{sup}) from the standard output layer, maximizing accuracy without interpretability constraints, and (2) interpretable BKT-logic predictions (p_{ref}) derived from student-specific grounded parame-

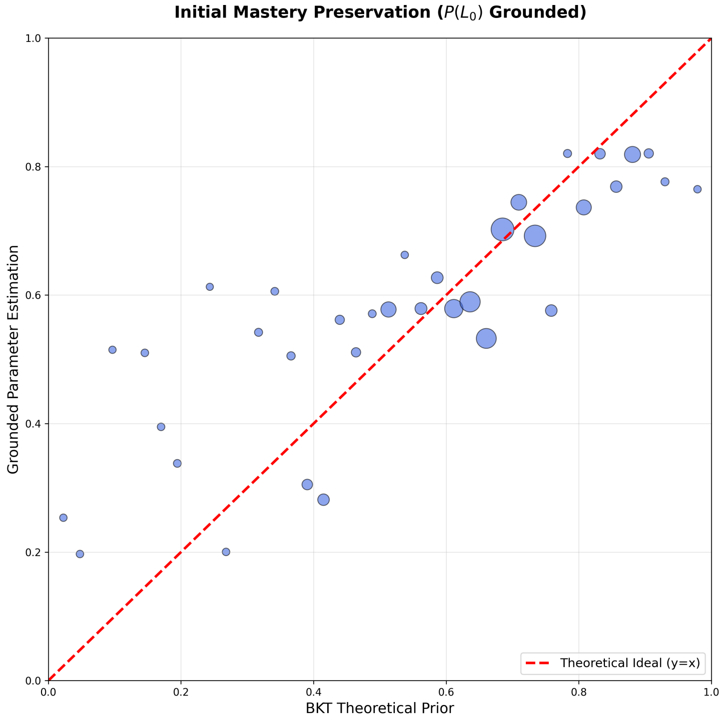

Figure 5. Semantic Alignment for Initial Mastery P_{L_0} (H1.2). Parity plot showing moderate monotonic relationship preservation (Spearman $\rho = 0.427$) between BKT theoretical priors and grounded parameters after neural transformation. Each bubble represents a binned aggregate of test interactions, with size encoding sample density. High-density regions (large bubbles) cluster near the diagonal (red dashed), demonstrating that rank-order preservation is robust where data is abundant while allowing individualization in sparse regions.

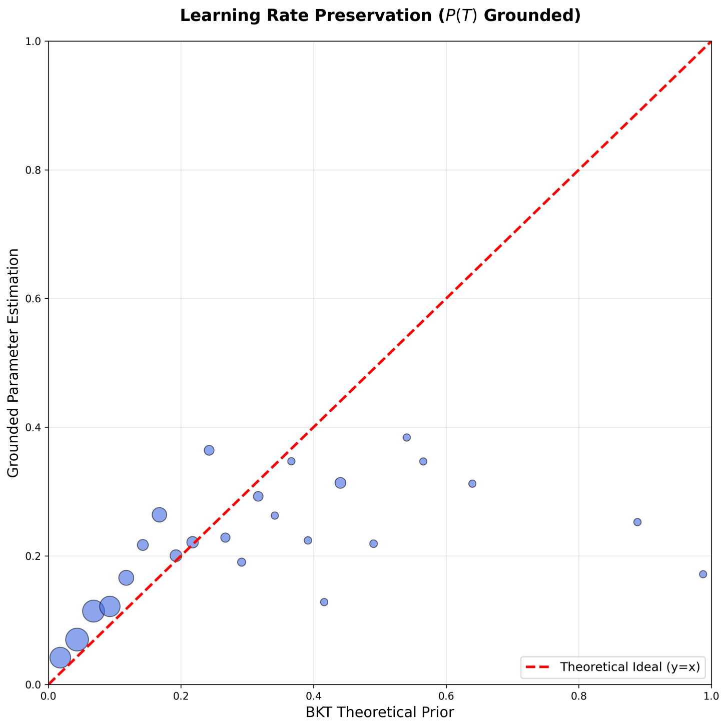

Figure 6. Semantic Alignment for Learning Rate P_T (H1.2). Parity plot demonstrating moderate-to-strong monotonic relationship preservation (Spearman $\rho = 0.604$) with superior alignment compared to P_{L_0}. The consistently tight clustering around the diagonal across the full P_T range indicates the model reliably preserves pedagogical understanding of practice effects through all neural processing layers while enabling context-aware refinement.

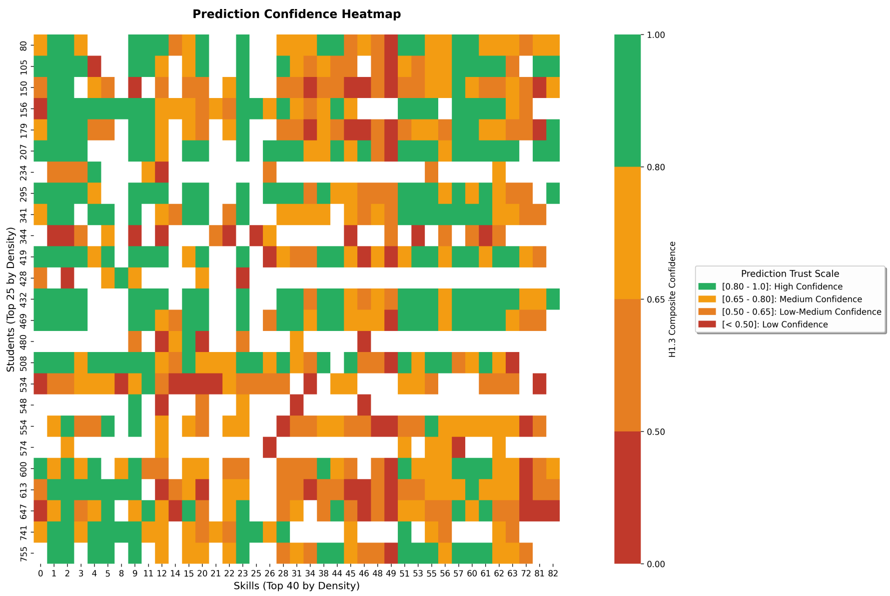

Figure 7. Student × Skill confidence heatmap for AS2009. Green cells indicate high confidence (≥ 0.8) where interpretable predictions can be trusted; yellow shows medium confidence requiring caution; red indicates low confidence pairs.

ters (p_{L_0}, p_T) extracted from the model's theoretical grounding layer, following classical BKT inference logic. For comparison, we include (3) classical BKT baseline (p_{bkt}) from a separate traditional BKT model using population-level parameters without neural individualization. Table 5 presents the test AUC results and the associated cost of interpretability ($p_{\text{sup}} - p_{\text{ref}}$).

Dataset	p_{sup}	p_{ref}	p_{bkt}	**Cost (%)**
AS2009	0.7824	0.6733	0.6097	0.1091 (13.9%)
AS2015	0.7073	0.6545	—	0.0528 (7.5%)
AL2005	0.8219	0.7361	0.7215	0.0858 (10.4%)
Bridge2006	0.8120	0.7025	0.6756	0.1095 (13.5%)
NIPS34	0.7991	0.6843	0.5729	0.1148 (14.4%)

Table 5. Accuracy–Interpretability Trade-Off Across Datasets. The table shows AUC scores for supervised predictions (p_{sup}), interpretable predictions (p_{ref}), and classical BKT baseline (p_{bkt}) using grounding mechanisms. Cost of Interpretability represents the performance gap between p_{sup} and p_{ref} (both from the same grounded model with ablation=none).

The empirical results confirm that gTransformer successfully bridges the gap between deep learning performance and theoretical interpretability. As shown in Table 5, the average *cost of interpretability* is 0.0952 AUC points (12.0% relative reduction), with the gap remaining remarkably consistent (7.4% to 14.4%) across all datasets. This validates our grounding architecture's ability to maintain high predictive power while strictly adhering to BKT causal logic. The gain from neural individualization is general: interpretable neural predictions outperform classical BKT in every comparable dataset, with improvements ranging from +2.0% (Algebra 2005) to +19.4% (NIPS Task 3/4). This confirms that capturing longitudinal behavioral patterns through individualized parameters provides a fundamental advantage over population-level models, even when constrained by interpretable logic.

The results reveal two insights regarding the accuracy–interpretability trade-off:

- **Consistent Low-to-Moderate Cost of Interpretability.** The average cost of interpretability across datasets is 0.0952 AUC points (12.0% relative reduction), demonstrating that theory-grounded interpretability can be achieved with remarkably low performance degradation. The cost is highly consistent across all datasets, ranging from a minimum of 7.4% (ASSISTments 2015) to a maximum of 14.4% (NIPS Task 3/4). This stability suggests that the neuro-symbolic framework is robust to varying dataset characteristics.
- **General Gain from gTransformer Individualization.** Comparing p_{ref} against p_{bkt} provides a measure of the *individualization advantage.* In all four comparable datasets, interpretable neural predictions consistently outperform classical BKT: ASSISTments 2009 (+10.4%), Algebra 2005 (+2.0%), Bridge to Algebra (+4.0%), and NIPS Task 3/4 (+19.4%). This consistent improvement demonstrates that gTransformer provides a general diagnostic advantage over population-level priors.

In summary, the cost of interpretability is consistently low and highly acceptable for practical educational applications. By accurately recovering student-specific learning parameters, gTransformer bridges the gap between black-box performance and theoretical rigor, offering a superior diagnostic alternative to traditional symbolic approaches without the high performance loss typically associated with constrained models.

5.4. Context-Aware Diagnostics (RQ3)

To validate RQ3, we show how gTransformer differentiates students based on their longitudinal learning context (their sequence of interactions) rather than just response patterns. This capability enables student-centered personalization beyond what population-based models such as BKT can provide. Classical BKT is fundamentally Markovian: given the same response sequence, it produces identical predictions regardless of the student's learning history. The gTransformer model, by contrast, uses its high capacity to capture temporal learning signatures (the individualized parameters p_{L_0} and p_T extracted from interaction history) to differentiate students even when their observable responses are identical.

The students were classified into four learning situations based on their historical learning parameters. Figure 8 shows students classified in the four quadrants for AS2009 dataset:

- **Low Prior Knowledge / Low Learning Rate**: Observations for students in this quadrant suggest limited foundational mastery with slow skill acquisition.
- **Low Prior Knowledge / High Learning Rate**: Despite initial knowledge gaps, rapid progress indicates responsiveness to instruction.
- **High Prior Knowledge / Low Learning Rate**: Strong foundational mastery but minimal progress may signal disengagement or instructional mismatch.
- **High Prior Knowledge / High Learning Rate**: Consistent mastery and rapid skill development suggest readiness for advanced material.

We selected representative skills with meaningful sequence length (5–30 interactions), high between-quadrant prediction range (strong differentiation), and students from at least two different quadrants with identical response sequences—the same answers to the same questions in the same order. These skills were used to build the 4×3 mosaic shown in Figure 9, where each subplot displays mastery predictions for students from different quadrants with identical response sequences within each skill. The dotted gray BKT lines overlap perfectly, reflecting the Markovian assumption: BKT produces identical probability estimates for all students who provide the same sequence of responses, regardless of their broader learning history. In contrast, the solid colored gTransformer lines diverge based on learning situation, demonstrating that the model attends to longitudinal context—

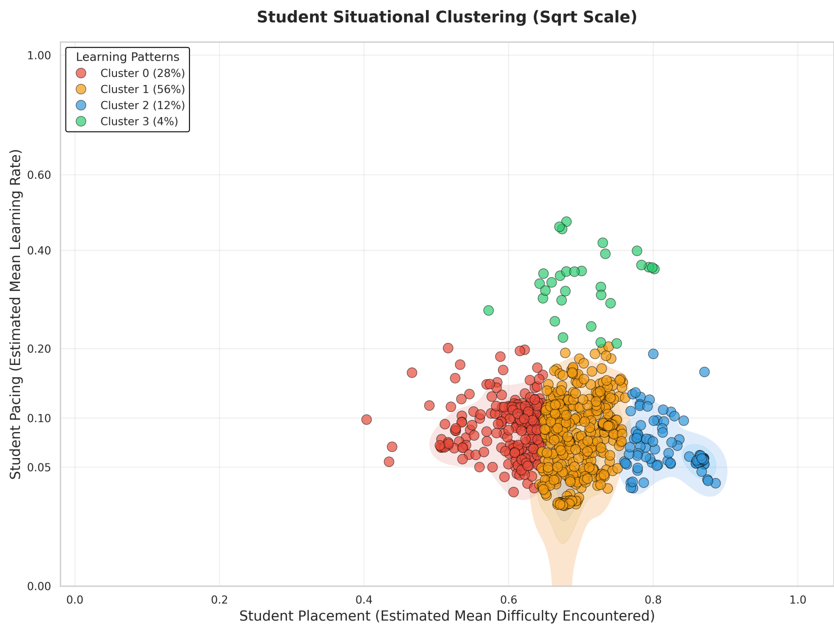

Figure 8. Student Clusters in AS2009
Students in AS2009 dataset are clustered into four learning situations based on their historical learning parameters (computed as averages across all previous interactions).

the full trajectory of previous interactions across all skills—to differentiate students even when their current skill-specific responses are identical. This visual evidence confirms that gTransformer captures non-Markovian personalization patterns that classical models fundamentally cannot represent.

This capability enables differentiated predictions for students with identical response sequences but different longitudinal learning contexts. For example, consider two students with the same sequence of responses to a given skill: one demonstrates rapid progress across previous skills (High P_{L_0}/High P_T), while the other shows gradual improvement through sustained practice (Low P_{L_0}/Low P_T). Classical BKT produces identical predictions for both students; gTransformer can generate distinct estimations reflecting their different learning trajectories. This differentiation may inform adaptive instructional decisions: educators could consider accelerated pacing for students showing rapid progress patterns, while providing additional scaffolding and practice opportunities for those demonstrating gradual development even when current skill performance appears equivalent.

The prediction divergence observed across different skill-sequence combinations shows gTransformer's practical value for student-centered personalization beyond traditional BKT. The model leverages its deep learning capacity to capture intricate interaction patterns such as learning history, temporal dynamics or individualized mastery trajectories, that Markovian models inherently cannot represent. This validates the hypothesis for RQ3: that *deep learning capacity + theoretical grounding = enhanced personalization*.

6. Conclusions

This work introduces gTransformer, a KT model that bridges the gap between deep learning performance and intrinsic interpretability through Latent Grounding. By anchoring Transformer latent representations to semantically meaningful constructs, we demonstrate that state-of-the-art predictive accuracy can coexist with theoretical alignment.

Our results validate the model's theoretical and practical utility across three dimensions. First, we confirmed that gTransformer internalizes BKT constructs as its primary organizing principle, achieving high structural selectivity and semantic alignment with

Figure 9. Context-aware skill mosaic showing non-Markovian personalization. Each panel displays students with identical response sequences (same ground truth bars) but different learning profiles (quadrants). BKT predictions (dotted gray) overlap due to Markovian constraints, while gTransformer predictions (solid colored) diverge based on historical learning parameters (P_{L_0}, P_T). This demonstrates that gTransformer differentiates students not by *what they answered*, but by *how they learned*.

theoretical priors ($RQ1$). Second, quantifying the accuracy–interpretability trade-off, we showed that interpretable neural predictions consistently surpass classical BKT (averaging +9.0% gain) with a low and consistent performance cost relative to black-box architectures ($RQ2$). Finally, we demonstrated that gTransformer enables context-aware personalization by capturing longitudinal learning patterns that Markovian models inherently overlook, allowing for differentiated diagnostics for students with identical scores but different interaction histories ($RQ3$).

Ultimately, gTransformer provides a practical model for transforming opaque deep learning predictors into actionable pedagogical tools. By grounding high-capacity neural networks in established domain theory, this approach offers a path toward student-centered personalization that is both highly accurate and pedagogically interpretable.

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Conflicts of Interest: The authors declare no conflicts of interest.

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